Python Programming – Project Report

Kinematic and Load Analysis of a Slider-Crank Mechanism for a Single-Cylinder Engine using Python

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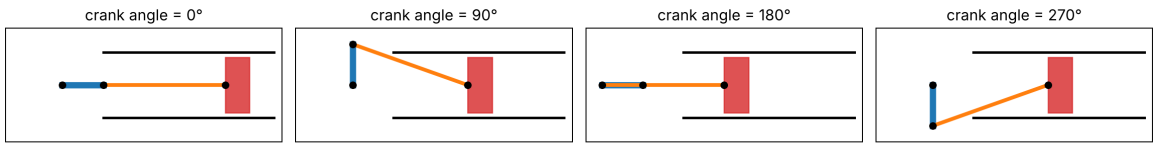


Figure 1. The slider-crank mechanism at four crank angles (TDC = 0°). Crank (blue), connecting rod (orange), piston (red).

1. Abstract

The slider-crank mechanism converts the reciprocating motion of a piston into the rotation of a crankshaft, and it sits inside almost every piston engine and compressor. This project builds a Python model of the mechanism for a single-cylinder, four-stroke engine and uses it to answer a design question: how do the piston motion, the connecting-rod force, the side thrust on the cylinder wall and the crank torque vary with crank angle, engine speed and rod length, and how large a flywheel does the torque curve require?

The exact kinematic equations are implemented with NumPy and compared against the two-term textbook approximation. An idealised gas-pressure cycle (polytropic compression and expansion, instantaneous combustion at top dead centre) is added to obtain forces and torque over the 720° cycle. An interactive Matplotlib simulator lets the user change speed, crank radius, rod length and piston mass. The model is checked in two independent ways: numerical differentiation of the position curve, and an energy balance between the torque integral and the p–V loop.

For an 80 mm bore, 100 mm stroke engine at 1500 rpm the model gives a peak gas force of 19.8 kN, a peak side thrust of 1.65 kN, a mean torque of 23.2 N·m (3.64 kW indicated) and a required flywheel inertia of 0.27 kg·m² for a speed fluctuation of 5 %. The study also shows that the reciprocating inertia is negligible at 1500 rpm but becomes comparable to the gas load at about 7352 rpm.

2. Problem Statement and Objectives

Problem. A designer choosing the geometry of a single-cylinder engine needs to know how the loads on the piston, rod, cylinder wall and crankshaft depend on the crank radius, rod length and speed. Doing this by hand at every crank angle is slow and error-prone, and the standard textbook formulas are approximations whose accuracy is rarely checked. A program can evaluate the exact equations at thousands of crank angles, vary the parameters, and plot the result.

Objectives

- Implement the exact position, velocity and acceleration of the piston as functions of crank angle.
- Measure how accurate the two-term approximation is, and how that depends on the ratio l/r.
- Add an idealised gas-pressure cycle and compute rod force, side thrust and crank torque over the full four-stroke cycle.
- Size a flywheel from the torque fluctuation.
- Study the effect of rod length and engine speed on the loads.
- Verify the model with checks that do not rely on the same equations.
- Provide an interactive simulator and a reproducible analysis script on GitHub.

3. Introduction and Background

In a slider-crank mechanism a crank of radius r rotates about a fixed centre, a connecting rod of length l joins the crank pin to a piston pin, and the piston is constrained to move along a straight line through the crank centre (Figure 1). When the piston is at its furthest position the crank angle is taken as 0° and the piston is at top dead centre (TDC); at 180° it is at bottom dead centre (BDC). The stroke is $2r$.

Two features make the mechanism interesting to analyse. First, the piston motion is not sinusoidal, because the rod swings through an angle φ as the crank turns. The piston is therefore faster in the half of the stroke nearest the crank and slower near the cylinder head. Second, the piston is a reciprocating mass: its acceleration is large (about 1.6 km/s² at 1500 rpm for this engine) and the force needed to produce it grows with the square of the speed.

In a four-stroke engine the crankshaft turns twice per cycle: intake, compression, power and exhaust strokes of 180° each. Only the power stroke produces positive torque, so a flywheel is used to keep the speed nearly constant. Machine-dynamics textbooks treat these topics with closed-form approximations [1, 2, 4]. Here the same physics is implemented numerically, which removes the need for the approximation and makes parameter studies fast.

4. Methodology and Algorithm

4.1 Kinematics

With the crank angle θ measured from TDC, the piston-pin distance from the crank centre is

$$x(\theta) = r\cos\theta + \sqrt{l^2 - r^2\sin^2\theta}$$

and the rod angle follows from the geometry

$$\sin\phi = \tfrac{r}{l}\sin\theta$$

For a constant crank speed ω, velocity and acceleration are obtained by the chain rule, $v = \omega\, dx/d\theta$ and $a = \omega^2\, d^2x/d\theta^2$. The program uses the exact derivatives of the expression above. The usual textbook approximation expands the square root and keeps two harmonics:

$$a \approx -r\omega^2\left[\cos\theta + \tfrac{r}{l}\cos 2\theta\right]$$

4.2 Cylinder volume and gas pressure

The piston distance below TDC is $(r + l) - x$, so the cylinder volume is

$$V(\theta) = V_c + A[(r + l) - x(\theta)], \quad V_s = 2rA, \quad V_c = \tfrac{V_s}{CR - 1}$$

The pressure over the 720° cycle is an idealised version of the air-standard Otto cycle. Intake is at constant pressure p_i. Compression is polytropic from BDC. Combustion is an instantaneous rise by the factor r_p at TDC. Expansion is polytropic and exhaust is at constant pressure:

$$p_{comp} = p_i\left(\tfrac{V_{BDC}}{V}\right)^n, \qquad p_{exp} = r_p\, p_i\, CR^{\,n}\left(\tfrac{V_c}{V}\right)^n$$

4.3 Forces and torque

The gas force acts on the top of the piston, with atmospheric pressure p_0 on the underside. The reciprocating mass m adds an inertia term $m{\cdot}a$, where a is the signed acceleration along the axis. Near TDC a is negative, so this term pulls the piston away from the crank and reduces the load that reaches it. The force transmitted along the cylinder axis is

$$F_{net} = (p - p_0)A + m\,a$$

Resolving along the connecting rod and across the cylinder gives the rod force R, the side thrust N on the cylinder wall, and the crank torque T:

$$R = \frac{F_{net}}{\cos\phi}, \qquad N = F_{net}\tan\phi, \qquad T = \frac{F_{net}\, r \sin(\theta + \phi)}{\cos\phi}$$

4.4 Flywheel sizing

Over one cycle the torque differs from its mean T_m. The running integral of $(T - T_m)$ over crank angle is the energy the flywheel must absorb and return. With the coefficient of speed fluctuation C_s the required inertia is

$$\Delta E = \max \int (T - T_m)\, d\theta - \min \int (T - T_m)\, d\theta, \qquad I = \frac{\Delta E}{C_s\, \omega^2}$$

4.5 Verification checks

Three checks are built into `analysis.py` so that the model is not judged only by how plausible its plots look:

- **Numerical differentiation.** The exact position is differentiated twice with `numpy.gradient` and compared with the analytic velocity and acceleration.
- **Energy balance.** The work per cycle found by integrating torque over crank angle must equal the indicated work from the p–V loop,

$$\int_0^{4\pi} T\, d\theta = \oint (p - p_0)\, dV$$

- **Two torque formulas.** The geometric expression for T above is compared with the virtual-work expression $T = -F_{net}\, dx/d\theta$.

4.6 Algorithm

1. Define the engine (bore, r, l, mass, compression ratio, polytropic index, pressures) in the `Engine` dataclass and reject geometries with l ≤ r.
2. Create an array of crank angles (0.05° steps over 720°).
3. Compute x, v, a and the rod angle φ for the whole array in one vectorised call.
4. Compute the cylinder volume and pressure for each stroke, then F_{gas}, F_{net}, R, N and T.
5. Run the verification checks and sweep l/r and engine speed by building modified engines.
6. Compute the mean torque and the flywheel inertia from the running integral of the torque.
7. Save figures, a CSV of the cycle and a JSON file of results; the simulator uses the same functions for its live display.

Table 1. *Engine data used in the analysis.*

Parameter	Value	Parameter	Value
Bore	80 mm	Compression ratio	8
Crank radius r (stroke 100 mm)	50 mm	Polytropic index n	1.3
Rod length l (l/r = 3)	150 mm	Intake / exhaust pressure	0.90 / 1.05 bar

Parameter	Value	Parameter	Value
Reciprocating mass m	0.5 kg	Pressure rise ratio r_p	3.0 at TDC
Swept volume	503 cm³	Atmospheric pressure	1.00 bar
Clearance volume	71.8 cm³	Base speed	1500 rpm

5. Python Implementation and Programming Concepts

The program is split into one module that contains the physics and three scripts that use it, so the simulator and the batch study can never disagree about the equations.

Table 2. *Files in the repository.*

File	Role
slider_crank_core.py	Kinematics, Engine dataclass, gas-pressure model, loads, flywheel sizing, work integral
slider_crank_simulator.py	Interactive animation with sliders and a pause button; plots x, v, a and shows inertia-only loads live
analysis.py	Runs the checks and the parameter studies; writes figures, CSV and JSON to outputs/
make_screenshots.py	Saves still images of the simulator window without opening a display
README.md, requirements.txt	Project description, run instructions, dependencies

Core kinematics function. It is written once and works on a single angle or on a whole NumPy array, in any consistent unit system:

```
def kinematics(theta, r, l, omega):
    """Exact piston position, velocity, acceleration for constant omega."""
    check_geometry(r, l)
    s_root = np.sqrt(l**2 - (r * np.sin(theta))**2)

    x = r * np.cos(theta) + s_root

    dx_dth = -r * np.sin(theta) - (r**2 * np.sin(theta) * np.cos(theta)) / s_root

    term_a = -r * np.cos(theta)
    term_b = -(r**2 * (np.cos(theta)**2 - np.sin(theta)**2)) / s_root
    term_c = -((r**2 * np.sin(theta) * np.cos(theta))**2) / s_root**3
    d2x_dth2 = term_a + term_b + term_c

    return x, dx_dth * omega, d2x_dth2 * omega**2
```

Programming concepts used

- **Functions and modules:** every calculation is a documented function in `slider_crank_core.py`, imported by the other scripts.
- **NumPy vectorisation:** a full 720° cycle (14 400 angles) is evaluated without a Python loop; `np.where` selects the pressure formula for each stroke.
- **Dataclasses:** the `Engine` class is a frozen dataclass with computed properties (area, swept volume, clearance); `dataclasses.replace` builds the modified engines for the l/r sweep.
- **Input validation and exceptions:** `check_geometry` raises `ValueError` if $l \le r$; the simulator shows a message instead of crashing when the sliders produce such a geometry.
- **Event-driven programming:** Matplotlib sliders and the button register callback functions; `FuncAnimation` calls `update` every 20 ms.

- **Dictionaries and lists:** `loads()` returns a dictionary of arrays; the l/r study builds a list of dictionaries, which is saved as JSON.
- **File input/output:** the `csv` and `json` modules write the results, and Matplotlib's non-interactive Agg backend writes the figures.
- **Numerical methods:** numerical differentiation (`np.gradient`), cumulative sums for integration, and a closed-loop work integral.

6. Results and Data Visualisations

6.1 Interactive simulator

Figures 2 and 3 show the simulator at two speeds. The mechanism is animated on the left. The right-hand plots show the exact piston position, velocity and acceleration for the current geometry, with a marker at the current crank angle. The text panel lists the inertia force, rod force and side thrust for the reciprocating mass alone.

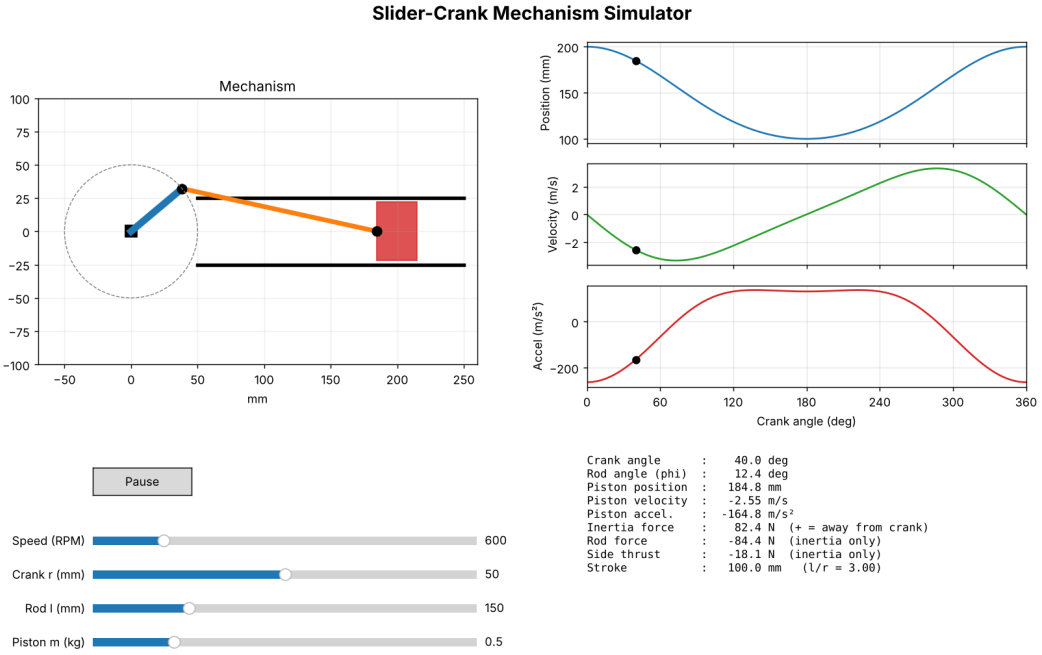


Figure 2. Simulator at 600 rpm, crank angle 40°.

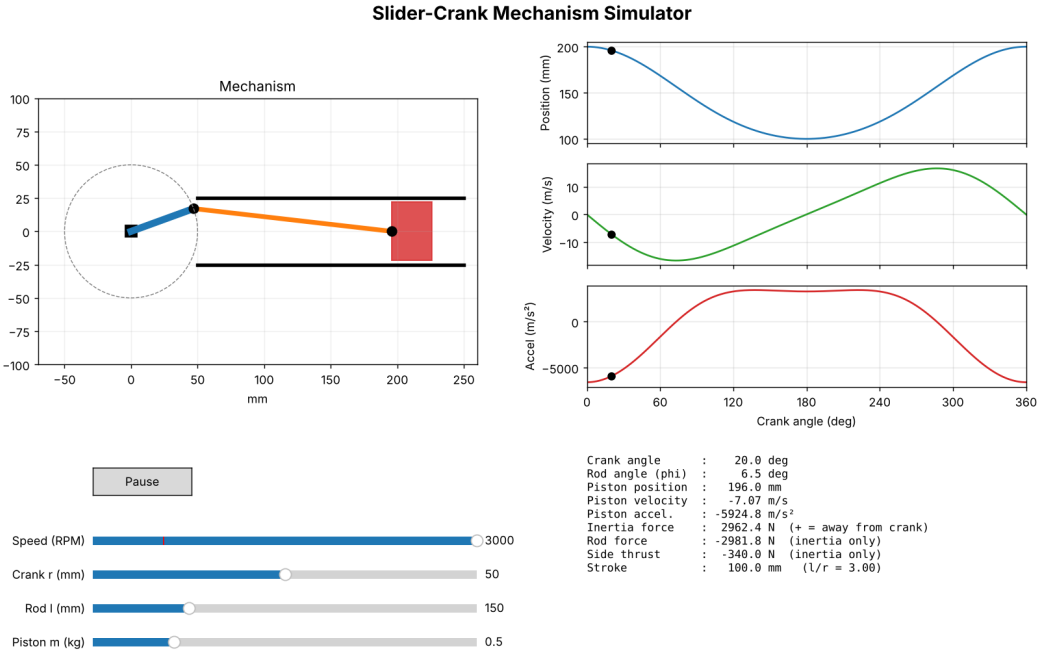


Figure 3. Simulator at 3000 rpm, crank angle 20°. The acceleration curve has the same shape as in Figure 2 but its scale is 25 times larger.

6.2 Piston kinematics and verification

At 1500 rpm the peak piston speed is 8.28 m/s, reached at a crank angle of about 287° (and symmetrically at 73°), while the mean piston speed is 5.0 m/s. The acceleration is 1645 m/s² towards the crank at TDC and 822 m/s² at BDC, matching the closed-form values $-r\omega^2(1 + r/l)$ and $r\omega^2(1 - r/l)$ (-1645 and 822 m/s²).

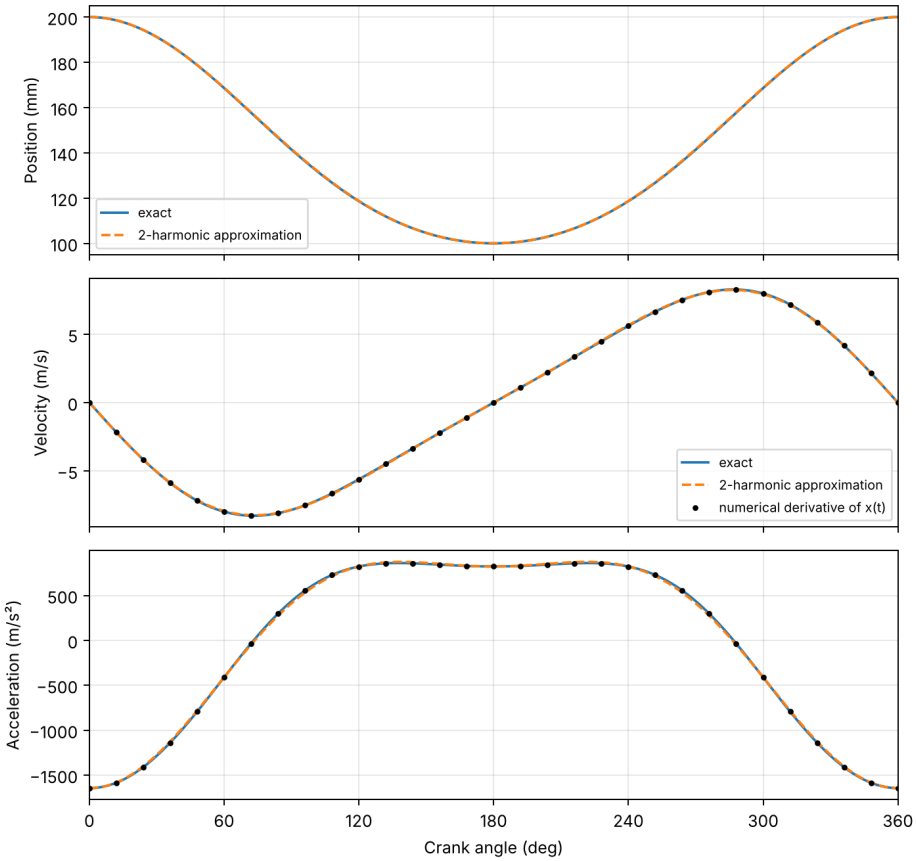


Figure 4. Exact and approximate piston position, velocity and acceleration at 1500 rpm. Black dots are the numerical derivatives of the position curve.

Table 3. *Verification results.*

Check	Result
Numerical derivative vs analytic velocity (max difference)	1.8e-05 %
Numerical 2nd derivative vs analytic acceleration (max difference)	4.2e-05 %
Two-term approximation vs exact acceleration (max difference, l/r = 3)	1.52 % (at 270°)
Work per cycle: $\int T\, d\theta$	290.97 J
Work per cycle: $\oint (p - p_0)\, dV$	290.97 J (difference 0.00002 %)
Geometric torque formula vs virtual work (max difference)	4e-13 N·m

6.3 Effect of the rod-length ratio l/r

Keeping the crank radius fixed and changing the rod length changes the approximation error, the peak acceleration, the maximum rod angle and the side thrust (Figure 5, Table 4).

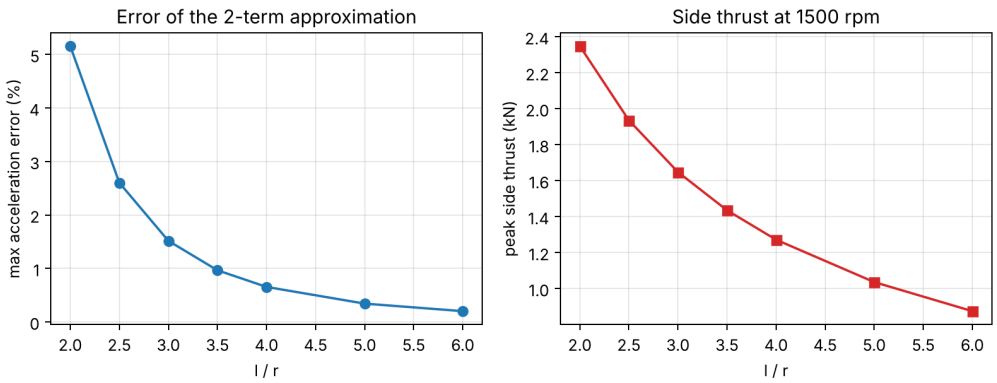


Figure 5. Left: error of the two-term acceleration formula against l/r. Right: peak side thrust at 1500 rpm against l/r.

Table 4. *Influence of l/r at 1500 rpm.*

l/r	Approx. error (%)	Peak accel. at TDC (m/s²)	Max rod angle (°)	Peak side thrust (kN)	Peak rod force (kN)
2.0	5.16	1851	30.0	2.35	18.83
2.5	2.60	1727	23.6	1.94	18.89
3.0	1.52	1645	19.5	1.65	18.94
3.5	0.97	1586	16.6	1.44	18.96
4.0	0.66	1542	14.5	1.27	18.99
5.0	0.34	1480	11.5	1.04	19.02
6.0	0.20	1439	9.6	0.87	19.04

6.4 Pressure and force over the four-stroke cycle

Figure 6 is the p–V diagram of the idealised cycle. The peak pressure is 40.3 bar and the area of the loop is the 291 J of work per cycle. Figure 7 shows the gas force, the reciprocating term and the net force. The net force peaks at 18.94 kN at the start of the power stroke, about 0.82 kN below the gas force because the inertia term is pulling the piston away from the crank at that instant. The minimum net force is -873 N (tension in the rod) at the TDC between exhaust and intake.

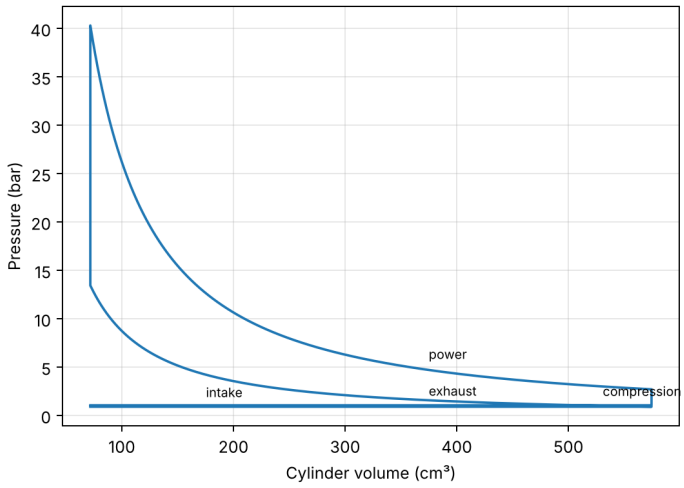


Figure 6. p–V diagram of the idealised cycle.

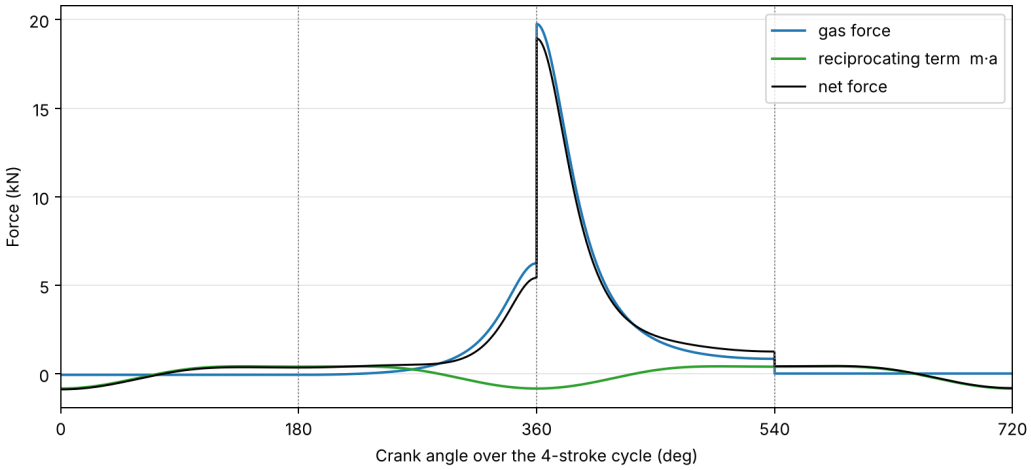


Figure 7. Gas force, reciprocating inertia term and net force at 1500 rpm.

6.5 Crank torque and flywheel

The torque (Figure 8) is positive and large only during the first part of the power stroke. It reaches 318 N·m, against a mean of 23.2 N·m, and is negative during compression, where the engine has to push the piston against the rising pressure. The mean torque times the angular speed gives an indicated power of 3.64 kW.

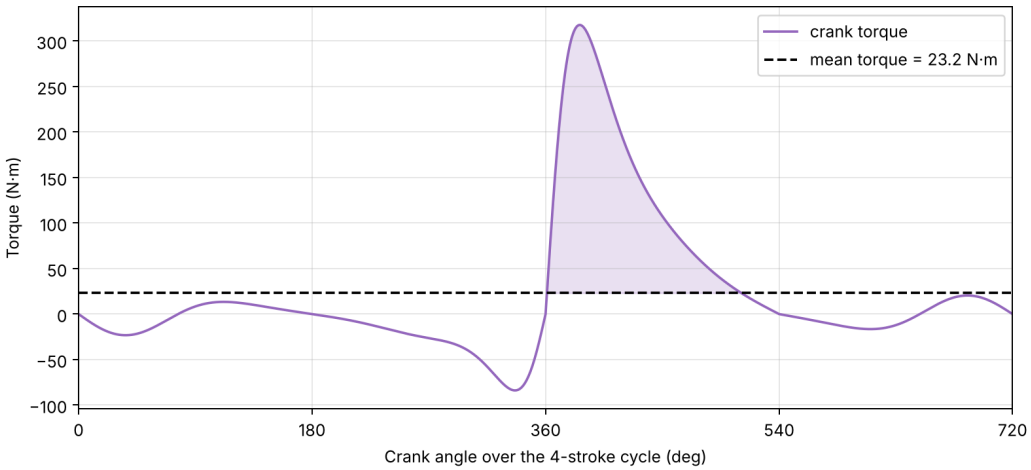


Figure 8. Crank torque over the four-stroke cycle at 1500 rpm.

Table 5. *Flywheel sized for a coefficient of speed fluctuation C_s = 0.05. The disc is steel (7850 kg/m³), 30 mm thick, treated as a solid cylinder.*

Speed (rpm)	Mean torque (N·m)	ΔE (J)	Inertia I (kg·m²)	Disc diameter (mm)	Disc mass (kg)
1000	23.2	332	0.606	402	29.9
1500	23.2	331	0.268	328	19.9
3000	23.2	328	0.067	232	9.9

6.6 Rod force and side thrust

The rod force is almost equal to the net force because φ stays small when the load is large. The side thrust is the component that presses the piston against the cylinder wall; it is largest, 1.65 kN, at a crank angle of 388° (about 28° after firing TDC), where a high gas force and a growing rod angle coincide. It changes sign when the rod swings to the other

side of the cylinder axis.

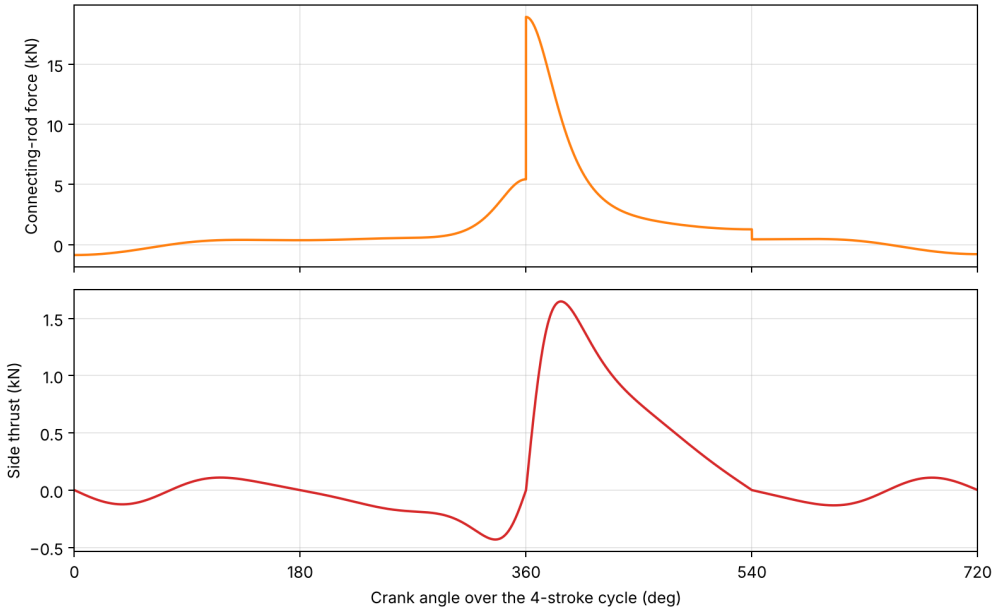


Figure 9. Connecting-rod force and side thrust at 1500 rpm.

6.7 Effect of engine speed

The peak reciprocating force at TDC is $m\,r\omega^2(1 + r/l)$, so it grows with the square of the speed (Figure 10, left). At 1500 rpm it is 822 N, 4 % of the peak gas force. It equals the peak gas force at about 7352 rpm for the 0.5 kg piston assembly, and at a lower speed for a heavier one. On the right, the tension in the rod at the TDC between exhaust and intake grows from 873 N at 1500 rpm to 3340 N at 3000 rpm and 13210 N at 6000 rpm.

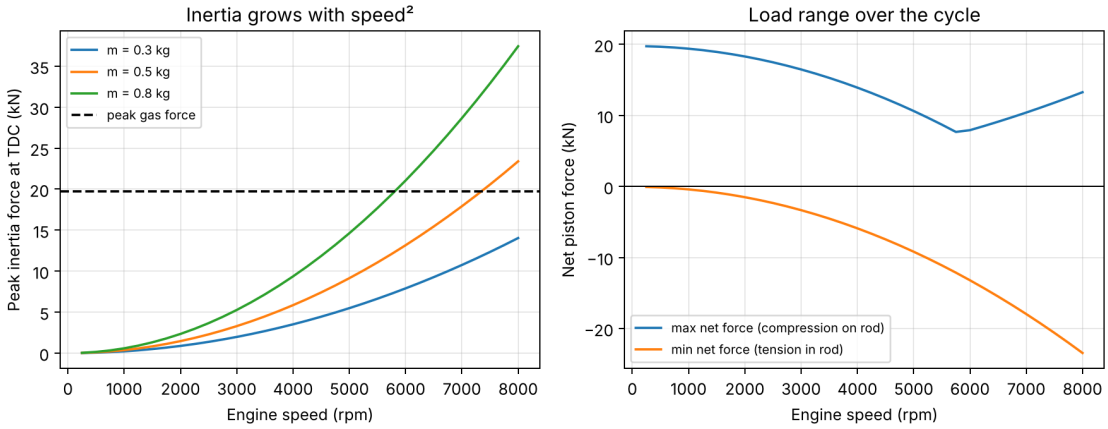


Figure 10. Left: peak inertia force at TDC against speed for three piston masses. Right: maximum and minimum net piston force over the cycle against speed (gas pressures held constant).

7. Discussion and Observations

The two-term approximation is exact at the dead centres. At θ = 0° and 180° the approximate and exact accelerations are identical, and the largest difference (1.5 % at l/r = 3) occurs near 90° and 270°, where the neglected higher harmonics act. The error falls quickly as l/r grows (5.2 % at l/r = 2, 0.2 % at 6), so the approximation is reasonable for ordinary engines but not for a short-rod, long-stroke design.

A longer rod reduces the side loads but costs size. Raising l/r from 2 to 6 reduces the peak side thrust from 2.35 to 0.87 kN and the peak acceleration at TDC from 1851 to 1439 m/s², while the peak rod force hardly changes because it is set by the gas load. The price is a taller, heavier engine, which is why practical engines sit around l/r = 3 to 4.

Inertia matters at speed, not at 1500 rpm. The reciprocating force is small compared with the gas load at the base speed, so the net force curve looks almost like the gas-force curve. In this model the tension in the rod at the exhaust/intake TDC exceeds the largest compressive load above about 5200 rpm (13 kN at 6000 rpm), so beyond that speed the inertia tension, not the combustion load, would set the design case for the piston pin, the rod and the rod bolts. This is the reason high-speed engines use light pistons and short strokes.

The kink in Figure 10 (right). The maximum net force falls with speed until about 5800 rpm and then rises again. Up to that speed the maximum is the firing load at TDC, reduced by inertia; beyond it, the largest compressive load on the rod is at BDC, where the inertia term is positive and grows with ω^2.

The torque is very uneven. The peak torque is about 14 times the mean, and for part of the cycle it is negative, so a single-cylinder engine needs a large flywheel: 0.27 kg·m² at 1500 rpm, which is a 328 mm steel disc of about 20 kg. The required inertia scales as $1/\omega^2$, so running the same engine at 3000 rpm needs a quarter of it, while ΔE barely changes (Table 5). Using several cylinders with staggered firing would spread the power strokes and reduce the flywheel.

Limitations. The model is intentionally simple. Combustion is instantaneous, which creates the sharp torque spike just after TDC and overstates the peak torque and side thrust; real pressure rises over some tens of degrees. The cylinder pressures do not depend on speed, friction and heat transfer are ignored, the rod mass is lumped into the reciprocating mass, the crank speed is held constant even though the flywheel calculation implies it fluctuates by 5 %, and balancing of the crankshaft is not considered. The numbers should therefore be read as trends and orders of magnitude for design comparison rather than as predictions for a real engine. The checks in Table 3 confirm that the equations are implemented correctly, not that the physical assumptions are realistic.

Possible extensions. Replace the idealised pressure with a measured or Wiebe-function combustion curve; add multiple cylinders with firing order; include rod mass split between the two ends; and add crank balancing.

8. Conclusion

A Python model of the slider-crank mechanism was built and verified, and it was used to study a single-cylinder four-stroke engine. The exact kinematics agree with numerical differentiation to better than 0.0001 %, and the torque integral matches the indicated work to better than 0.001 %. The two-term textbook approximation is accurate to about 1.5 % at l/r = 3 and is exact at the dead centres. The parameter studies show that rod length trades side thrust against engine size, that reciprocating inertia is negligible at 1500 rpm but sets the peak rod tension above roughly 5200 rpm, and that the strongly pulsating torque of a single cylinder requires a flywheel of about 0.27 kg·m² at 1500 rpm. The interactive simulator and the analysis script, both built on one shared physics module, are available in the GitHub repository so that every figure and table in this report can be regenerated.

9. References

[1] J. J. Uicker Jr., G. R. Pennock and J. E. Shigley, *Theory of Machines and Mechanisms*, 5th ed., Oxford University Press, 2017.

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